Candidate Pack Summary Requested by System Talentlink on 22/09/2025.

Candidate Pack - George Jenkinson

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Lecturer in Assistive Robotic Technologies (ACAD108256)

- 6.ACAD PIF
- 7.ACAD additional questions (Questionnaire)
- Cover_letter_AP_AR_SEMT_GJ.pdf (Attached Cover Letter)
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- Lec_Rob_Teaching_Statement.pdf (Other attachment)
- Lec_Rob_Research_Statement.pdf (Other attachment)

6.ACAD PIF, by George Jenkinson, 22/09/2025

Title	Dr
First Name	George
Last Name	Jenkinson
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Main contact telephone number	+44 7482400417
Address Line 1	40 Daisy Road
Address line 2	Easton
Town/City	Bristol
Postcode	BS5 6JS
Country	United Kingdom, South West

Which of the following options describes your right to work status in the UK?

I am a UK or Irish national with a current Passport

Do you have any unspent or pending criminal convictions?

no

What is your desired work pattern?	Full time

Desired annual salary (for example £25,000 as 25000)

43482

Current contractual notice period	Immediately available

7.ACAD additional questions, by George Jenkinson, 22/09/2025

Are you a recent University of Bristol graduate (within the last 12 months)?

No

Are you currently working with UoB (either as an employee or through an agency)?

Yes

Were you referred to this role by a current UoB employee?

Yes

Where did you first hear about this role?

From a UoB employee

Are you a member of any relevant professional body (for example RCGP, ACCA, APM, BCS)?

Professional body	IEEE
Date achieved	-
Registration number	-
Registration expiry	-

Fitness to practice proceedings (Clinical academic vacancies only)

-

Please provide contact details for three professional referees. If you do not have three professional referees you can provide personal referees. Your referees will automatically be contacted should you be shortlisted for interview

References

Reference

Title/Form of address	Dr
First name	Antonia
Last name	Tzemanaki
Email	antonia.tzemanaki@bristol.ac.uk
Daytime Phone Number	-

Reference

Title/Form of address	Prof
First name	Andrew
Last name	Conn
Email	a.conn@bristol.ac.uk
Daytime Phone Number	-

Reference

Title/Form of address	Dr
First name	Xavier
Last name	Aure-Calvet
Email	xavi.aurecalvet@uwe.ac.uk
Daytime Phone Number	-

Please confirm that you have provided three referees

Yes

Do you consent that we may contact your referees?

Yes

Professor Andrew Conn
School of Engineering Mathematics and Technology
University Of Bristol
Ada Lovelace Building
Bristol
BS8 1TW

George Jenkinson
40 Daisy Road
Easton
Bristol
BS5 6JS
07482400417

For 21st September 2025
RE: Lecturer in Assistive Robotics

Dear Professor Andrew Conn,

I have been proud to be part of SEMT in its previous forms as a Masters and PhD student, and now as a Teaching Associate and Senior Research Associate due to its tendency toward collaborative projects across a broad set of subdisciplines, and the quality of the commitment by staff to support research and learning. Over this time, I have been supported in developing my own research avenues, and I have expanded my teaching and related administerial experience significantly over the last year as a Teaching Associate. With experience in teaching and supervising at many levels over 9 years, and research focussed on both foundational and applied areas with a particular recent focus on using technology to improve quality of life and equity for visually impaired people, I believe that my experience and ethos mean that I meet all of the specifications set out in the job description, making me a strong candidate for the position advertised.

In this document I broadly cover how I meet the job specification. My teaching and research statements cover the relevant aspects much more thoroughly. In those documents I lay out my research vision, with particular emphasis on how I plan to expand my body of research in new directions, exploiting my existing international and interdisciplinary network. I connect my ideas for fundamental research to tangible expected outputs that will have significant social and academic impact. My teaching statement focuses on an inclusive, student-centred ethos, where I discuss my past experience and ideas for innovation in the teaching space, with particular focus on how AI is ushering in a new teaching paradigm, and how we can respond to this and turn it into a tool for the benefit of teachers and students. Excerpts with particular relevance in those documents are **emboldened**.

Research Record

The publication list in my CV demonstrates my broad and strengthening research record, with many IEEE publications amongst others in swarm, soft, HRI, assistive, and medical subdisciplines of robotics, as well as in using AI for accessibility purposes in the humanities. Within the last year I have won AHRC funding to create an AI-powered workflow to help curators and museums make their collections more accessible for visually impaired audiences, and EPSRC funding to develop a prototype intended to help Braille-reading skill acquisition. As part of preparing these bids and executing the research, I reached out and successfully engaged industry stakeholders as well as charity organisations who have helped organise user studies.

Teaching at University Level

I have taught undergraduate students since being a second year undergraduate in Amsterdam with my own registered tutoring business, where more recently I have also been teaching and supervising masters students as part of the SEMT teaching staff. This broad teaching experience for my career stage has given me valuable experience in teaching students from different cultural and academic backgrounds, as well as students with diverse learning styles and needs. As a result, I have a good understanding of how cultural differences can influence learning, and am practiced in approaching learning material from multiple approaches in order to meet students' needs. Tutoring students requires lateral thinking in order to break down concepts and particular problems into manageable blocks that they may rebuild for themselves. The optimal approach may differ significantly between particular students, and the creativity required to address this and reconceptualise my own understanding is something that I find stimulating and rewarding.

This experience was valuable when I took on the role of teaching assistant for undergraduate and master's level courses at Bristol during my PhD. COVID-related restrictions moved classes into online environments, and interactive resources such as discussion forums and online meetings became indispensable to deliver support around resource material. I was particularly aware during this time that the students I would meet with in small online meetings were having very limited contact with the university, and so I provided a point of contact for them. This was especially important considering the logistical, physical and emotional stresses that many students were under due to the circumstances imposed by the pandemic.

Industry Experience

Alongside my MSc in swarm robotics at the University of Bristol, I worked as a production technician at Open Bionics, where I gained valuable insight into the industry-led approaches that positively impact people's quality of life. The ethos driving the company is to design affordable, bespoke, simple-to-use assistive devices, leading to design with significant input from end-users. The main product of the company uses sEMG devices to control a prosthetic arm. Fabricating and testing these prosthetics gave me practical experience with bio-interfacing robotic devices, techniques relevant to rapid prototyping, and knowledge of standards around assistive technologies in broader public use. I maintain frequent contact with many members of staff where we often discuss mechatronics and assistive devices.

I trust that the submitted documents demonstrate that my ongoing research, broad published record on technology for improving quality of life and health equity, and experience and enthusiasm for teaching, make me a good fit for the role, and show that I am capable of meeting the responsibilities required. I would enjoy the opportunity to build my expertise, expand my experience, and realise the next steps of my research vision in this role. I believe that I would contribute positively and productively towards the school from the offset if appointed.

Thank you for your consideration,

Dr. George Jenkinson

GEORGE P. JENKINSON

+447482400417 georgejenkinson1@gmail.com

Google Scholar

EDUCATION

Mechanical Engineering PhD, Bristol Robotics Laboratory, University of Bristol — Oct 2019 - Feb 2024

Thesis Title: *Soft Variable Stiffness Tactile Sensors for Clinical Examinations*

Robotics MSc, Distinction, University of Bristol and University West of England — Sep 2018 - Aug 2019

Thesis Title: *A Bio-Inspired Mechanism for Dynamic Role Allocation in Robotic Swarms: Self-Organised Personality Dynamics for Swarm-Size Sensitive Risk-Taking*

Liberal Arts & Sciences BSc, Cum Laude, University College Amsterdam — Feb 2015 - Feb 2018

Thesis Title: A Novel Holistic Analysis Method for the Mechanical Behaviour of Giant Unilamellar Vesicles

Major in Physics - **Thesis of Distinction Award** (Soft Matter Biophysics)

RELEVANT EXPERIENCE

Research Associate - Centre For Print Research, UWE — *Bristol, UK*, March 2024 - present

Teaching Associate – University of Bristol, School of Engineering Maths and Technology — *Bristol, UK*, Dec 2024 – Sep 2025

Graduate Teacher – University of Bristol — *Bristol, UK*, Sep 2020 - Feb 2022

Open Bionics – *Mechatronic Prosthetics*. Production Technician — *Bristol, UK*. Aug 2018 - Apr 2019

Private Tutor – Self employed — *Amsterdam, NL*. Sep 2016-Apr 2019

AMOLF – *Soft Biological Matter Group*. Research Intern — *Amsterdam, NL*. Jan - Dec 2017

PUBLICATIONS

My first author publications range over swarm robotics, soft robotic sensors, soft locomotion robotics, medical robotics, assistive robotics, machine vision, qualitative public engagement, and generative AI.

Jenkinson GP, Samaroudi M, & Aure Calvet X. *Tactile Embroidery Reproduction Exploiting Machine Vision for Visually Impaired Engagement*. In *Digital Heritage 2025* (pp. 1-10). Eurographics. https://doi.org/10.2312/dh.20253349

Jenkinson GP, Aure Calvet X. *Matching the Stitch: Autonomous Classification of Embroidery Techniques for Replication* Textile Conservation Conference @ 50, Glasgow, United Kingdom, 2025 Jun

Aure Calvet X, & **Jenkinson GP**. *Beyond the display case: New methods for documenting and accessing historic textile collections*. Textile Conservation Conference @ 50, Glasgow, United Kingdom, 2025 Jun

Jenkinson GP, Conn AT, Tzemanaki A. *Coil-Tac: Coiled Capacitor Mechanism with Liquid Metal for Tactile Sensing*. *IEEE Robotics and Automation Letters*, vol. 10, no. 5, pp. 5194-5200, May 2025 https://doi.org/10.1109/LRA.2025.3558655

Jenkinson GP, Conn AT, Tzemanaki A. *Braille-tip: Structured Small-Footprint Tactile Sensor for High Acuity Dynamic Tactile Tasks*. 2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob), Heidelberg, Germany, 2024, Sep. https://doi.org/10.1109/BioRob60516.2024.10719951 Youtube

Jenkinson GP, Tiemann K, Papathanasiou A, Bewley J, Conn AT, Tzemanaki A. (2023) *A robotIc Radial palpatIon mechaniSm for breast examination (IRIS)*. 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Busan, Korea, Republic of. 2023 Aug. https://doi.org/10.1109/RO-MAN57019.2023.10309324. Youtube

Wharton P*, You TL*, **Jenkinson GP**, Diteesawat RS, Le NH, Hall EC, Garrad M, Conn AT, Rossiter J. (2023). *Tetraflex: A Multigait Soft Robot for Object Transportation in Confined Environments*. IEEE Robotics and Automation Letters, *8*(8), 5007–5014. doi: 10.1109/LRA.2023.3290409

Jenkinson GP*, Houghton N*, van Zalk N, Waller J, Bello F, Tzemanaki A. (2023). "Acceptability of Automated Robotic Clinical Breast Examination: Survey Study". *Journal of Participatory Medicine https://doi.org/10.2196/42704*

Jenkinson GP, Conn AT, Tzemanaki A. (2023). "ESPRESS.0: Eustachian Tube-Inspired Tactile Sensor Exploiting Pneumatics for Range Extension and SenSitivity Tuning". *MDPI Sensors*, *23*, 567. https://doi.org/10.3390/s23020567. YouTube

Jenkinson GP, Conn AT, Tzemanaki A. (2020). "A Pressure Controlled Membrane Mechanism for Optimising Haptic Sensing" *Towards Autonomous Robotic Systems - 21st Annual Conference, TAROS 2020, Nottingham, UK, Proceedings - TAROS*. https://doi.org/10.1007/978-3-030-63486-5_31. YouTube

Jenkinson GP*, Hunt ER*, Wilsher MJG, Dettmann CP, Hauert S. (2020). "SPIDER: a Bioinspired Swarm Algorithm for

Adaptive Risk-Taking" *ALIFE 2020: The 2020 Conference on Artificial Life.* https://doi.org/10.1162/isal_a_00279

Bewley J, **Jenkinson GP**, Tzemanaki A. (2021). "Optical-Tactile Sensor for Lump Detection Using Pneumatic Control" *Frontiers in Robotics and AI*, 8(July), 1–12. https://doi.org/10.3389/frobt.2021.672315. YouTube

Jenkinson GP. (2016). "Newton the Natural Philosopher Does Newtonian Physics carry Newton's God?" *AUC Undergraduate Journal of Liberal Arts & Sciences, 56.* Link

Media Coverage - Please see my webpage link for embedded media including:

BBC Radio Bristol (2023 and 2024), BBC News (2023 and 2024), The Independent, Somewhere on Earth podcast (2023, 2024), ITV News, and more.

Funded Project

Research Council

Matching the Stitch October 2024 – March 2025. Funding body: AHRC. Value: ~£5,000. Description: Creating an AI based workflow to restore embroidered artworks with true-to-original colouring and texture.

Braille-pen October 2025 – February 2025. Funding body: EPSRC. Value ~£20,000. Description: Developing Braille-tip prototype into a marketable product and investigating market opportunities.

Industry

Digitally Enhanced Artefacts September 2025. Funding from: SS Great Britain Museum. Value: £3,000. Description: Fine-tuning generative AI models to restore lost details for digital display in museum.

Tactile Car Models from Heritage Photos October 2025. Funding from: British Motor Museum. Value: £5,000. Description: Generating digital and physical 3D models of cars from archive photographs for tactile interaction.

Digitisation of Engineers' Drawings October 2025. Funding from: The Building Archives. Value: £15,000. Description: Developing machine vision models to extract details from archival hand drawn plans

ACADEMIC PRESENTATIONS & AWARDS

Talk - *Coil-Tac: Coiled Capacitor Mechanism with Liquid Metal for Tactile Sensing* *IEEE IROS, October 2025*

Talk - *Tactile Embroidery Reproduction Exploiting Machine Vision for Visually Impaired Engagement* *Digital Heritage,* September 2025

Talk - *Matching the Stitch: Autonomous Classification of Embroidery Techniques for Replication* *Texcons@50,* June 2025

Talk – *Harnessing Generative AI Models for Heritage Restoration* *Nat Trust Textile Conservation Studio*, Feburary 2025

Talk - *From Soft Robotics to Tactile Interfaces* *CFPR lunchtime seminars,* July 2024

Workshop Host – Making Exhibition Spaces Accessible for the Visually Impaired October 2024

TakeAIM award Shortlist – Smith Institute Award for Mathematics in Healthcare March 2024

Talk - *Braille-tip: Structured Small-Footprint Tactile Sensor* *IEEE BioRob*, September 2024

Talk - A robotIc Radial palpatIon mechaniSm for breast examination (IRIS) *IEEE RO-MAN*, August 2023

Invited Speaker - The ARTEMIS project: using robotics and soft sensors to detect early-stage breast cancer *Interdisciplinary Research Symposium on the Prevention, Early Detection and Treatment of Cancer*, January 2023

Poster - A Soft Tactile Sensor for Tasks Requiring High Spatial Resolution *IROS*, October 2022

Poster - Exploiting Mammalian Inner Ear Mechanics for Increased Pressure Sensitivity and Range *Robosoft*, April 2022

Host - Co-chair for 1 of 4 tracks *BRL Conference*, June 2021

Talk - SPIDER: a Bioinspired Swarm Algorithm for Adaptive Risk-Taking *ALIFE*, July 2020

Award - Thesis of Distinction February 2018

Invited Speaker - When More is Different. Fundamental Emergence in Entangled Quantum States

International Interdisciplinary Seminar - Differentiating humans from animals and machines, January 2017

SKILLS

Technical Skills - Competent writing and using code in python, lisp, Matlab, and C++.

Working knowledge of PHP, html, C, Java and related programming languages. Experienced with comfyui, Diffusion based image generation, Segmentation, fine-tuning open-source AI models, LLMs, Solidworks, Matlab, Arduino IDE, Tensorflow, Weka, StableDiffusion, opencv.

Practical experience with 3D printers, CNC milling, laser cutting, soldering electronics, robotic arms, elastomeric moulding in a Soft Robotics Laboratory, and a Biohazard Level 1 Laboratory.

Soft Skills –

Communication and working in a team

I have frequently collaborated on projects internationally, across institutions, within lab groups, and have led groups of MSc students and undergraduates in research projects. Recently, I have mediated the relationship between MSc students and industry partners for the students' end of year projects. This included carefully and clearly communicating the expectations and requirements of the partners from the university, as well as ensuring that the students were able to construct projects based on their own ideas whilst benefitting from the guidance of the industry partners.

I have worked across institutions to allocate and organise marking regimes for the inter-institutional Robotics MSc between Univeristy of Bristol and University of the West of England. This required significant communication around which staff members had the most relevant expertise, and negotiating fair workloads across teams and in response to individual requests.

Time management

I have held jobs alongside my undergraduate and post-graduate studies, as well as undertaken a large building project on my home during my PhD. I have managed to hold research and teaching roles across institutions simultaneously at times, requiring fore-planning and careful allocation of time to meet the demands of both. Balancing these undertakings and meeting deadlines demonstrates my ability to manage time effectively.

Ability to work independently

Particularly exemplified during the pandemic, when I made a "home-lab" from which I carried out a large amount of the experimental work behind the ESPRESS.0 paper and the paper led by Bewley. These used a customised 3D printer, home made vacuum chamber and syringe driver, as well as code to interface between the custom devices.

REFERENCES

Xavi Aure-Calvet
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Centre For Print Research
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Andrew Conn
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University of Bristol
Associate Professor of Robotics
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Antonia Tzemanaki
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Telephone available on request

Teaching Statement

Values

My teaching is guided by the values of **inclusion, equality, student-focussed delivery, and prioritising student wellbeing**. I believe these values combine to make students feel individually valued which enhances their learning, engagement, and overall experience.

Experience

I have taught in some capacity for **9 years**. Notably, whilst an undergraduate, this included writing and delivering a 20 unit course over 2 years which was recognised and accredited by Clonlara School as equivalent to AS-level in Physics, Maths, Chemistry, and Biology. Much of this process is comparable to writing a university level course, **identifying quantifiable and achievable learning objectives** that combine over modules to make a rounded learning experience.

During my PhD, I co-supervised 3 individual MSc projects and 1 group project, 2 of which led to successful publications in a good journal and conference. This year I supervised 17 individual MSc student projects across the three robotics disciplines and Data Science, of which I consider there to be a strong chance of publishing 3. The projects included 3 data science industry partnerships which meant I acted in a role new to me as a university representative and often authoritative go-between for students and the partner. I am currently collecting feedback, and hope to establish a standard for feeding back from industry partners to the school, as currently there appears to be no mechanism in place to collect or record this.

In a supervisory role, it is especially straightforward to be guided by my teaching values, which imply a bespoke approach that is considerate of the individual. I was pleased that during their vivas, given time to make an open statement to augment their presentations, a significant number of my supervisees chose to use the time to thank me for a particular contribution or act as mentor over the project period. Gaining feedback from students is valuable and can be difficult to obtain, almost always biased to represent those with stronger opinions.

Factors of good teaching

I have outlined below what I consider to be three of the most important and relevant factors that broadly answer the question "What makes teaching good?", including a brief discussion of the challenges of each, and an approach to address these challenges.

Innovation

At the edge of a paradigm shift it is important to evaluate what is important about teaching. Widely available AI models have invalidated many long-standing examination and learning processes in the same way that electronic calculators invalidated arithmetic. The product of demonstrating learning has become comparable, or in many cases inferior, to what can be produced automatically with little effort. However, teaching and learning is about **development of critical skills**. In this way, learning to **synthesise knowledge**, **solve problems creatively**, and **appraise critically** are timeless core skills that students need to develop. A major challenge of teaching is to encourage students to value these over the end-product of their efforts.

Teaching methods such as **challenge based learning** develop these skills in a way that is compatible with a new paradigm. Project-based learning that rewards collaboration and exploitation of a broad range of skills to address a real-world need develops all of the core skills identified above. Such projects can be "open book", which students find liberating and rewarding as the link to skills useful outside of academia is obvious.

Assessments across universities have always shifted in response to technological development. As opposed to shoe-horning assessment methods onto modern students by constructing synthetic environments with no access to the latest technological developments, careful assessment design that **acknowledges and incorporates the latest developments** offers more value to students[1], and is more appealing for potential employers. Importantly, in cases where restrictions are beneficial for the learning of the student, this must be communicated in a way that the students understand and agree with.

[1] HEPI - Student Generative AI Survey 2025

I believe there is latent potential to enhance personalised learning for student-facing and teacher-facing education interfaces[2]. On many courses within SEMT, the beginnings of such personalisation is taking form. For example, I am helping to create revision material covering parts of preliminary courses that many students have fed back that they struggle with. With efficient data collecting and well organised material, identifying students for which this is most appropriate, or identifying areas where most students have a deficit, and taking steps to improve should be straightforward. Careful and **fair incorporation of emerging AI technologies should enhance all sides of education**, accounting for and addressing disparities in attitudes to using AI across student demographics[1].

Inclusion and Equality

For students, a university is first and foremost a place of learning. Universities are powerful institutions that hold significant weight in public discourse. University graduates are over-represented in positions of power across the board. From this perspective it is a notable failure that the intake of universities does not come close to reflecting the demographics of country or city-wide population in the typical university age bracket.

Whilst this failure does not fall squarely with universities, I am pleased that the University of Bristol and particularly SEMT actively takes action to attempt to address this deficit with initiatives that include targeted visits to local schools, affirmative action in job advertisements, and has a strong presence of groups such as Women in Robotics to enhance the university's sometimes impressive history of inclusivity.

Whilst studying, working, and teaching in Bristol and abroad, I have seen and benefited from the **value in culturally diverse cohorts** in terms of learning approaches and cultural exchange. SEMT celebrates an international intake that adds significant value for its students and staff.

Teaching diverse student cohorts necessitates careful consideration of material and delivery. I subscribe to the idea that difficulties in teaching due to different cultural backgrounds or ways of thinking constitutes a **double empathy problem**, where potential difficulties are mutual and bidirectional, rather than being a deficit on the side of the learner. I find it useful to consciously and deliberately apply this framework to any teaching in order to distil the important learning points and break down any teaching aid into its constituent parts. This is valuable for students with different cultural reference points and learning preferences. My experience tutoring international students with significant neurodivergences as part of my tutoring business in Amsterdam was enlightening with respect to seeing the **benefit of providing multiple avenues of engagement with material**. This approach makes teaching accessible and fair without diminishing its effectiveness.

Feedback

It is supremely important to respond to the successes and failures of course delivery. The weakness in most student feedback systems is that from the students' perspective, feedback is charitable and the reward is intangible, meaning that feedback is scarce and has significant responder bias.

To close the loop, students need to see a response to their own feedback. **Adaptive units that respond to a student's history and feedback are possible** with new technology and careful planning. Data exists to identify where a student may have a consistent weakness over units, but is passing overall. For example, gaming the exam system may mean that it is more beneficial for them to focus on mastering some course material, rather than generalising over all of it, meaning that higher level units that build on foundational skills present a significant barrier for them. Students may also feed back that a section of the course to date has been too brief. These two examples can be prepared for such that the delivery of material on a course is responsive and bespoke, key aspects of closing the feedback loop on the side of the student as well as the course provider.

Plans for Teaching

The trio of Robotics, Biorobotics and Aerial robotics Masters offers students with diverse but compatible interests a way to develop their skills in a potentially new discipline to graduate at an expert level. The diverse intake is at once a strength and a weakness of the courses, where the spread of skills that students arrive with means that collaborative projects can yield significant peer-to-peer learning, yet starting on different pegging with respect to many skills can make meaningful assessment of their learning over a unit difficult. **Incorporating techniques that are able to respond adaptively and curate a**

[2] ChatGPT: The end of traditional assessments in higher education?

collection of bespoke learning materials specific for a student's needs is a means to begin to address this difficulty. I believe this is possible as part of a new paradigm that exploits the latest technological developments particularly in AI, and so would look to create a sandbox for exploring this possibility with unit directors, particularly those who teach units that feed directly into each other, e.g. Engineering Maths 1 and 2.

Across Robotics, Biorobotics, and Aerial Robotics, the range of units offered by SEMT is broad and comprehensive. I have and would continue to draw from my experience across soft, tactile, medical, HRI, assistive, and swarm robotics, knowledge of machine vision and AI systems, and background in Physics to contribute positively to many of them. I believe there is space to develop a unit around robots as research devices. This would be complementary to a number of existing units whilst not overlapping them. The explicit purpose of the robotic systems studied in this unit would be those designed to enable research. This is opposed to robots that are designed for an end-use. Examples include biomimicry when used to study biological systems, and cognitive robotic systems, social robotics, and machine vision when used to study people and behaviour.

Research Statement

Research Direction

I am primarily interested in **robotics as a way to investigate the human experience**; how the material world is interpreted and processed by our senses to become a subjective reality.

Technological and scientific breakthroughs move the goalposts on what is considered to be sentient, intelligent, subjective. This has been informative in, for example, neuroscience, where neural correlates have provided a mechanistic backbone for a number of behaviourally diagnosed conditions. This has allowed for the development of a number of clinical practices and led to a maturing of discussions around mental health in broader society. Robotics and AI contribute to this discourse by offering an avenue for scientific investigation, a methodology to execute research across disciplines, and a means to create technology that performs complex biological or biomechanical functions. This simultaneously tests and demonstrates our best working models for understanding these functions, whilst broadly benefitting society.

I believe that robotics can contribute significantly to the investigation of how our senses work, and can be a means to investigate and develop experimental protocols across disciplines in a way that makes it uniquely and powerfully interdisciplinary.

Current Assistive Research

Investigation into human function and experience dovetails with making assistive robotic devices that generally perform all or part of a human function in a complementary role. This is exemplified in my ongoing work using machine vision as a means to interpret artworks for visually impaired audiences. The academic investigation focuses on whether physically consistent and technically feasible textiles emerge as a by-product of highly fine-tuned image generating AI models. This **implicitly mimics the visuo-tactile experience** of looking at a highly textured surface, and so lends itself to interpreting embroidered artworks as a sighted person is able, for visually impaired audiences. I have published articles this year on the cataloguing and matching of embroidery techniques by generative AI models, and the production pipeline and evaluation by end-user study of creating tactile replicas for visually impaired audiences to access heritage goods. A journal article is in preparation for the ability of generative AI to implicitly learn technically feasible embroidery from a training set of images. I believe that using synthetic systems to study what can be inferred from minimal information is fundamental to investigating the nature of human senses and experiences. This approach reveals the core elements of sensory perception and allows us to explore these phenomena in a tangible way, even within an artificial framework.

A significant focus of my PhD was on tactile sensors, driven by the desire to understand the minimal information necessary to recreate aspects of touch. Stemming from research into how a decaying time component of touch could be coupled with a known trajectory for surface texture analysis, my research led me to investigate Braille as a stimulus for soft tactile sensors. This proved to be a charismatic application and led to widespread news coverage including BBC radio Bristol as well as forming the basis for a successful EPSRC Exploratory Award bid on which I am co-I and working as a senior research associate from October.

Factors of Impactful Research

Outlined below are 3 features of research that contribute significantly to its worth. They are frequently given as significant factors considered by funding bodies. For each I have given examples of the standard that my current research meets, and my plans to exceed this in the future.

Interdisciplinary, Multi-Institutional Collaboration

I have worked across disciplines with a number of universities, charities, and industry partners now and in the past. **Collaborative projects have clear and demonstrable benefits** in improving the quality of research output, breadth of impact, and in follow-on projects. Diverse perspectives from other institutions as well as across industry **add significant academic value and improve impact**, perhaps by consideration of a feasible path to commercialisation.

For my current and recent collaborative projects see Table 1. This includes funding bids that are advanced and/or in the process of being written. Together these demonstrate an ability to form **suc-**

cessful collaborations with industrial and academic partners nationally and internationally and across institutions and disciplines.

I have found initiating and developing good collaborative partnerships to follow easily after an exciting idea is established. I intend to keep developing my research network and actively seek to do so at research events, as well as directly reaching out to potential collaborators, and responding to those who reach out to me.

Dissemination

Academic and public engagement with research is valuable for feedback, testing feasibility, and increasing interest in research projects. I have attended and spoken at a number of workshops, seminars, symposia, and conferences, but consider dissemination in general media for the public to be most important for projects which concern them. For me, this has included appearances on BBC Radio Bristol, local BBC and ITV news, as well as their national websites and a number of other media outlets regarding my research around autonomous robotic breast health examinations and a Braille learning assistive device.

I have been a return guest to BBC Radio Bristol as well as some podcasts and have given interviews to the same newspapers for different projects. These repeated invitations and engagements demonstrate a good working relationship and that I am a known entity to these media outlets. I would look to maintain good relationships with media outlets as well as look to form new ones when projects are ready to be publicly disseminated.

Academic and Social Impact

Collaboration and dissemination both contribute to impact. The key factor in increasing impact is that the core of the research addresses a relevant problem or shortcoming in a broader societal context. I believe that connecting foundational research to applications that can change people's lives, or influence policy, is the most important quality by which to assess research. Foundational research may be hugely impactful indirectly, where multiple applications and avenues of future research are based on it.

Near-Future Research Plans

I have advanced project plans including partners in neuroscience at the University of Birmingham and internationally across 2 institutions in Sweden: the University of Gottenburg and the University of Linkoping, to develop an experimental robotic testing platform using microneurography to investigate the neural response to affectionate touch. This will be posed with the purpose of exploring how to use a wearable to enhance this neural response. A certain type of neurodivergency linked to autism spectrum behaviour is also linked with a diminished neural response to affectionate touch. A later stage of research is intended to investigate whether the stimuli that enhance the neural response in neurotypical

Project	Funding Type	Description	Partner(s)	Role	Continuity at UoB?
Digitally Enhanced Artefacts	Industry	Fine-tuning generative AI models to "restore" lost details for digital display in museum	SS Great Britain Museum	Co-I	No
Tactile Car Models from Heritage Photos	Industry	Generating digital and physical 3D models of cars from archive photographs for tactile interaction	British Motor Museum	Co-I	No
Digitising Engineering Drawings	Industry	Developing machine vision models to extract details from archival hand drawn plans	The Building Archives	Co-I	No
Matching the Stitch	Research Council (AHRC)	Training generative AI to produce technically feasible embroidery images	Holburne Museum	Co-I	Finished (Apr 2025)
Braille-Pen	Research Council (EPSRC)	Development of Braille-learning assistive device prototype and exploration of commercialisation	Bristol Braille Technology LLC	Co-I	Yes, at UoB
Robotics for Affective Touch	Research Council	Development of robotic platform to investigate neural response to social/affective touch	Universities of Gottenburg; Linkoping; Birmingham	(PI)	Yes, currently in writing stage
ACCESS Music: Inclusive Music Literacy Platform for the Visually Impaired	Research Council (Horizon Europe)	Development of a musical notation device with dynamic Braille output and synchronized auditory feedback.	Universities of Edinburgh; Cambridge; Glasgow; Oxford; Oslo; York; University College Dublin; Western Norway University of Applied Sciences	Technical Advisor	Yes, currently in writing stage

Table 1: Table of recent and ongoing collaborations that I have been involved with from the bid-writing stage.

people has the same effect on neurodivergent people, possibly **enhancing our understanding of some of the neural mechanisms linked to some types of neurodivergence**. An immediate step upon appointment would be to apply for a Research Development International Collaboration Award (RDICA) to support a **short international visit and workshop to prepare for a larger funding bid**, for example an EPSRC New Investigator Award.

I would seek to develop this foundational line of using robotics to investigate the properties that mediate human senses and experience, and develop applied technologies based on these for assistive purposes. This would feed into the writing of a taught unit with a focus on robotics for interdisciplinary research, rather than directly for application, as detailed in my teaching statement.

Summary

Assistive robotics can offer a clear connection between foundational research and application as I have demonstrated in my research and discussed here. I have outlined the broad vision and first steps I would like to follow given the opportunity to expand my research on investigating the human experience using robotics, and applying the developments to assistive technology. My research would aim to positively impact people's lives with assistive technologies and expand our understanding of how our senses and internal models interact with the material world to produce experiences.